Bad Blood V2: Attention-Based Population Models and Reference-Conditioned Personalization

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***Abstract*—This report consolidates the trained models and completed evaluations of Bad Blood V2. A compact residual one-dimensional convolutional network with final-stage squeeze-and-excitation (CNN+SE) estimates systolic and diastolic blood pressure (SBP/DBP) from ten-second physiological signal windows. Matched ECG+PPG and PPG-only versions are retained. Source training uses 390,600 segments from 1,085 participants in the VitalDB-derived partition of PulseDB. Three-seed main-test participant-macro MAE is 12.39/8.00 mmHg for ECG+PPG and 12.32/7.85 for PPG-only. Reference-conditioned personal adaptation learns a pressure change relative to known early personal measurements. With matched calibration data, both versions improve upon personal mean in both pressures for every seed in the six-person and sixteen-person target cohorts. Dual-input personalization also improves upon bias correction for both pressures in every seed in the six-person cohort, reaching 8.32/3.76 mmHg. That joint advantage does not fully transfer to the sixteen-person cohort, where bias correction remains stronger for DBP. We provide architecture definitions, fitting details, complete error metrics, controls, checkpoint files and runnable inference.**

***Index Terms*—blood pressure, PPG, ECG, squeeze-and-excitation, calibration, personalization**

I. Release decisions and contribution

The engineering objective is to obtain compact population and personalized estimators that can use newly observed physiological signals. This report records the *20261013 final V2 snapshot*, not a new training run or a claim that every scientific hypothesis succeeded. It contains two reusable outcomes.

Population model. We retain high-resolution residual CNN+SE with 38,296 parameters in two input configurations. ECG+PPG is the primary configuration within this family because its three-seed validation score and pressure-range stress-test result are better. PPG-only is a useful single-sensor alternative with slightly lower main-test MAE. The three-seed architecture-wide validation comparison originally selected the plain CNN, whereas SE had the lowest observed test MAE among its six required architecture candidates. The retained SE research model should not be described as winning both rankings.

Personal model. We retain PairSE, a reference-conditioned pressure-change model built on the same CNN+SE. Both modalities beat personal mean across all three seeds and both target cohorts. Dual-input PairSE provides the clearest six-person evidence of simultaneous improvement over mean and bias correction; PPG-only has lower personal-development error and slightly lower average sixteen-person PairSE error. There is no demonstrated universal input-mode winner. Both learned models and both simple controls remain in the package.

TABLE I
Fixed population partitions after participant exclusions.

Partition	Participants	Segments
Training	1,085	390,600
Validation	192	69,120
Main test	141	56,400
AAMI-designated stress subset	116	666

II. Data, task and information boundaries

PulseDB combines cleaned ECG, PPG and invasive arterial pressure signals from VitalDB and MIMIC-III [1]. Our experiments use its *VitalDB-derived population partition and individual-record files*, not the whole combined collection. Each model input is a synchronized ten-second window sampled at 125 Hz; each output is one SBP and one DBP in mmHg. The label comes from the arterial-pressure reference. ABP samples, current BP labels, demographics and participant IDs are not population prediction inputs.

The corrected population split excludes historically explored personal subjects; the subsequent personal cohort is also checked for disjointness. The AAMI subset is evaluated separately and is not added to the main-test count as an independent experiment. Its name refers to the dataset construction, not certification of our estimator. Previously evaluated test cohorts are retained for controlled comparison; they are not presented as untouched external tests.

Personalization uses four development participants to choose settings and two target cohorts: the original six and a subsequent sixteen. The original six provide 4,284 later query windows; the sixteen provide 2,310. The sixteen were unseen at their first evaluation but had already been examined before this modality study. Cohorts are therefore reported separately, not combined into an apparently new validation population. The support is all eligible windows in the first 20% of elapsed recording time, with a ten-second guard before later query windows. Gaps in the recording clock are retained. These dense public-data references are not actual cuff events or ND measurements.

TABLE II
CNN+SE ARCHITECTURE; SHAPES OMIT BATCH DIMENSION.

Stage	Operation and output
Input	2 channels × 1,250 samples; no initial decimation
Stem	Conv1d 2→8, kernel 7, stride 2, pad 3; BN; SiLU. 8×625
Block1	Residual 8→16, stride 2. 16×313
Block2	Residual 16→32, stride 2. 32×157
Block3	Residual 32→48, stride 2. 48×79
SE gate	Temporal mean; Linear 48→6; ReLU; Linear 6→48; sigmoid; multiply channels
Readout	Global temporal mean; dropout 0.15; Linear 48→32; SiLU; Linear 32→2

III. POPULATION ARCHITECTURE AND TRAINING

A. Signal representation and network

The supplied filtered ECG and PPG fields are scaled independently within each window to the unit interval. The tensor shape is $N \times 2 \times 1250$, ordered ECG then PPG. The PPG-only fit is trained from scratch without ECG. An internal mask zeros ECG in both ordinary forward inference and reference encoding. Keeping the two-slot representation maintains a controlled parameter-matched comparison; it is not a claim that ECG is needed for the PPG model.

Each residual block has a kernel 7 convolution, BN and SiLU, followed by a kernel 5 convolution and BN. A stride-matched kernel 1 projection and BN form the skip branch; SiLU follows the sum. The SE mechanism [2] reweights learned feature channels after the final residual stage. These channels mix signal features; their weights are not direct sensor importance scores. CBAM and temporal-only attention were also compared using the same backbone. Adding SE increases parameters from 37,666 to 38,296, about 1.7%.

B. Objective, optimizer and checkpoint selection

For normalization $N(\cdot)$, the physical-unit population prediction is

$$p(X) = \mu + \sigma \odot f_\theta(N(X)), \tag{1}$$

where training-only label center $\mu = (115.41765, 62.92810)$ and scale $\sigma = (18.94188, 12.08322)$ are stored in the checkpoint. The objective is mean squared error in standardized SBP/DBP, not a mixture of test metrics. Both modes use AdamW, learning rate 0.001, weight decay 0.001, batch 256, input noise SD 0.003 and gradient-norm clipping at 5. A plateau scheduler halves the learning rate after three validation non-improvements. Maximum training is 40 epochs, with early-stopping patience 8. Checkpoints minimize the mean of SBP and DBP participant-macro validation MAE. All six SE fits stopped after nine epochs and retained epoch 1 (Table III); later epochs were evaluated but did not improve that criterion. Seeds 42,43,44 quantify training variation. Seed 42 is the representative within each modality because its validation score is best, not because of a test-set ranking.

TABLE III
POPULATION FITS AND VALIDATION-SELECTED CHECKPOINTS. BOLD: LOWEST VALIDATION SCORE PER INPUT CONFIGURATION.

Input	Seed	Run epochs	Best epoch	Val. score
dual	42	9	1	**10.291**
dual	43	9	1	10.457
dual	44	9	1	10.421
ppg	42	9	1	**10.374**
ppg	43	9	1	10.583
ppg	44	9	1	10.406

TABLE IV
DUAL-INPUT ARCHITECTURES: THREE-SEED PARTICIPANT-MACRO MAE (SBP / DBP), MMHG. BOLD: LOWEST PER PRESSURE AND SPLIT.

Model	Parameters	Validation	Main test
Plain CNN	37666	**12.41** / 8.33	12.59 / 8.36
CNN+SE	38296	12.51 / **8.26**	**12.39** / **8.00**
CNN+temporal	37680	12.49 / 8.32	12.58 / 8.31
CNN+CBAM	38310	12.52 / 8.29	12.42 / 8.11
CNN+additive	38296	12.44 / 8.34	12.66 / 8.18
CNN+SE-all	38670	12.47 / 8.32	12.49 / 8.23

IV. REFERENCE-CONDITIONED PERSONALIZATION

A. Model definition

Directly fitting all population parameters on a small personal reference bank did not reliably improve the simple controls. PAIRSE instead makes known reference BP an explicit input and learns waveform-dependent pressure changes. This follows the general reference-conditioning and layer-selective transfer ideas in prior work [3], [4], implemented here with our matched CNN+SE and chronological evaluation.

The shared encoder produces $h_c, h_r \in \mathbb{R}^{48}$ and standardized population predictions $p_c, p_r \in \mathbb{R}^2$ for current and reference windows. Features are standardized using source-training feature statistics. With known standardized reference pressure y_r, define

$$z_{cr} = [h_c - h_r,\ h_r,\ y_r,\ p_c - p_r], \tag{2}$$

$$z_{rr} = [0,\ h_r,\ y_r,\ 0], \tag{3}$$

$$\widehat{y}_{c|r} = y_r + (p_c - p_r) + g_\phi(z_{cr}) - g_\phi(z_{rr}). \tag{4}$$

The change network g_ϕ is a 100→64→32→2 MLP with SiLU between layers and 8,610 parameters. Its last layer starts at zero during source training. The full shared-encoder model has 46,906 parameters; personal fitting updates only the 8,610 change-head parameters. The subtraction anchors identical current/reference waveforms to their known reference pressure. Final inference averages $\widehat{y}_{c|r}$ over *all* early references and converts back to mmHg. Current or later reference BP is never passed to inference.

B. Source pair training and personal fitting

Source pair-head training freezes the CNN+SE and uses all 390,600 source training windows. A random non-self window from the same source participant serves as each reference; source subset timestamps are not recoverable, so this stage is not a source chronological forecasting experiment. The head

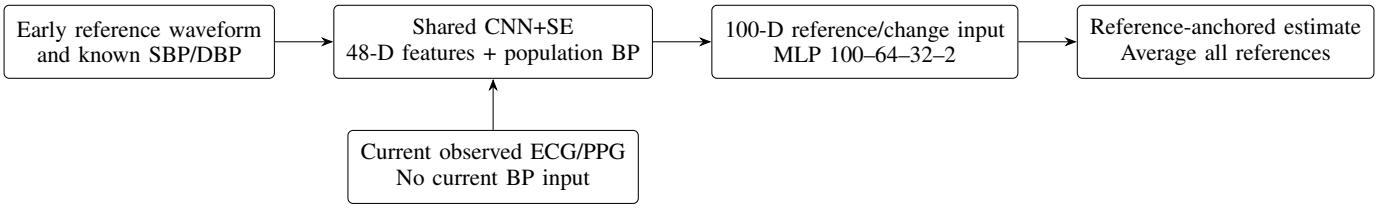


Fig. 1. PAIRSE inference. The reference bank belongs to the same person and precedes the evaluated query period. Source-trained representation and personalized change head are shared across all reference/query pairs.

uses Adam lr 0.001, batch 2,048, standardized MSE, clipping 5, up to 20 epochs and validation patience 5. Source feature centering/scaling is fixed thereafter.

Personal development compares no adaptation and eight combinations of scope (head-only/full), learning rate(0.0003/0.001) and steps(25/100). It minimizes the worst of the four development MAE ratios to personal mean and bias correction, with mean MAE as tie-breaker. Settings are chosen independently for each modality, averaging seeds on the same four development people; both choose **head-only Adam, lr 0.0003,25 steps**. This selection does not assume every development baseline is beaten: dual development SBP is 7.53 versus 6.76 for bias correction; PPG development is 7.23/4.30 versus 9.17/5.43 for bias correction and 12.53/5.26 for personal mean.

All eligible early support windows participate in at least one time-separated pair. Each fitting step uses all valid chronological support pairs, and only support labels enter the loss. BatchNorm running statistics and dropout stay fixed; there is no personal weight decay or L2-SP penalty. The frozen encoder choice is empirical, not an imposed requirement to keep personalization small. Later query labels are used only for evaluation, not stopping or tuning.

C. Fair controls

Every method receives the same reference bank S_i and query set Q_i for participant i. The personal mean needs no query waveform:

$$\widehat{y}_{iq}^{\text{mean}} = |S_i|^{-1} \sum_{r \in S_i} y_{ir}. \tag{5}$$

The stronger additive control uses the same modality's pretrained predictor:

$$\widehat{y}_{iq}^{\text{bias}} = p(X_{iq}) + |S_i|^{-1} \sum_{r \in S_i} \{y_{ir} - p(X_{ir})\}. \tag{6}$$

Bias correction already tracks the population model's within-person changes; it is not equivalent to predicting a constant. PAIRSE must improve the learned change relationship, not simply recover the person's typical BP.

V. METRICS AND COMPARISONS

Let $e_{ij} = \widehat{y}_{ij} - y_{ij}$. We report MAE, RMSE, signed mean error (Bias), sample error SD with denominator $n-1$, and fractions with $|e| \le 5, 10, 15$ mmHg. Participant-macro metrics compute each person's metric and then weight people equally. Pooled metrics treat every window equally. Macro SD averages within-person SD and excludes some between-person offsets; it must not be substituted for pooled SD. Each pressure is evaluated separately.

All main tables average the metrics of three separately trained/evaluated seeds, not an ensemble's predictions. Paired confidence intervals resample people after averaging each person's matched-seed difference. Repeated seeds are not independent participants. Per-person MAE, tail errors, pressure strata, correlations and R^2 are also retained in the machine-readable results.

A. Population results

Table IV provides the architecture comparison. Plain CNN test MAE is 12.59/8.36; SE is 12.39/8.00 and CBAM is 12.42/8.11. Plain CNN has the best mean validation score 10.371, narrowly ahead of SE10.390; therefore the test improvement is an observed comparison, not validation proof of SE dominance. The new matched modality study uses the same three seeds and source data for both SE models. Dual validation MAE is 12.51/8.26; PPG-only is 12.50/8.41. Main-test PPG-only MAE is 12.32/7.85, but its signed bias is more negative. On the AAMI-designated subset, macro MAE is 19.99/12.62 for dual and 20.80/13.19 for PPG. This motivates retaining both input modes and using dual as the population default rather than declaring PPG universally better. Table V provides all requested pooled metrics.

B. Where personalization beats both controls

On the original six people, dual PAIRSE gives 8.32/3.76 versus 9.47/4.22 for mean and 9.16/4.40 for bias correction. The relative improvements over bias correction are 9.2% for SBP and 14.6% for DBP. Each of the three seeds improves both endpoints against both controls. Seed-averaged per-person improvement over bias correction occurs in five of six people for each endpoint. Descriptive paired MAE-difference 95% intervals are $[-1.46, -0.23]$ for SBP and $[-0.95, -0.29]$ for DBP. The joint primary-MAE criterion is therefore met on this cohort.

The PPG version also improves the six-person average to 8.19/3.85, but its SBP advantage over PPG bias correction is only 0.08 mmHg and appears in one of three seeds. Some threshold metrics do not improve even where MAE does; the complete tables preserve these differences.

C. Expanded-cohort result

On sixteen people, dual PAIRSE reaches 12.67/7.61 and PPG PAIRSE reaches 12.47/7.51, both better than mean 17.41/9.71. This improvement holds for SBP and DBP in every seed. However, the own-modality bias controls reach 12.82/7.48 (dual) and 12.74/6.89 (PPG). PPG PAIRSE improves macro SBP in all three seeds but worsens DBP in all three. Dual improves SBP in two of three and DBP in one of three. Table VIII shows every seed.

For PPG, paired mean-baseline differences are$-4.95/-2.20$ mmHg, with 95% intervals$[-8.09, -1.52]$ and$[-3.53, -0.92]$. Against bias correction, differences are$-0.27/+0.63$, with intervals$[-2.12, 1.53]$ and$[-0.04, 1.26]$. PPG-versus-dual PAIRSE intervals cross zero for both pressures. Neither input modality establishes a stable overall winner in this cohort.

Pooled results also matter. PPG PAIRSE reduces pooled Bias from 2.72/1.92 to$-0.25/0.37$ relative to bias correction, but pooled SD increases from 14.49/8.03 to 15.28/9.15 and MAE increases from 12.03/6.58 to 12.34/7.30. The small macro SBP gain reverses when weighting windows rather than people. Lower aggregate signed bias alone is not evidence of lower prediction error.

D. Follow-up calibration and retained method

The completed residual follow-up evaluated twenty centered linear/kernel calibration configurations and support-only forward validation, using the same references and three seeds. Settings were selected on the four development people before target evaluation. The dual selection used a feature RBF residual correction for SBP and prediction-feature linear ridge for DBP; PPG selection retained the previous PAIRSE for both outputs. The new dual sixteen-person MAE is 12.34/8.23, versus 12.67/7.61 for PAIRSE; its six-person result is 9.87/4.43 versus 8.32/3.76. Better SBP does not satisfy the joint criterion when DBP worsens. We therefore retain PAIRSE and archive the full negative follow-up rather than silently changing one endpoint after inspecting target results.

VI. IMPLEMENTATION AND REPRODUCIBILITY

The GitHub final-results folder separates `01_pretrained` and `02_personalization`. It contains six population checkpoints, six source pair-head checkpoints and 132 participant-specific checkpoints ($22 \times 3 \times 2$). The shared implementation includes the exact backbone, modality mask and pair architecture, a portable inference loader and the fixed personal adaptation routine. Source snapshots and full-study instructions preserve the original fitting procedures. Checkpoints include label normalization statistics; PairSE also stores feature normalization.

For a new participant, start from that modality's source PairSE checkpoint, fit only on early known references, and supply the same reference bank during later prediction. A different person's fitted checkpoint is not a generic deployable personal model. PPG inference may accept $N \times 1 \times 1250$; the wrapper inserts the zero ECG slot. Raw, unfiltered arbitrary sensor data are not accepted as though already matched to the training representation.

The experiment verification covered 816 units and replayed 39,564 target predictions exactly in the recorded environment. The residual follow-up verified 1,260 units with maximum replay difference1.53×10^{-5} mmHg. Release checks load every packaged checkpoint, compare the portable loader with the original inference, and check the selected personal fit and ECG-invariance of the PPG path. These checks concern software correctness, not new generalization evidence. Git records, artifact SHA256 hashes, complete per-person metrics and the editable report are included.

The population model is compact (about 150 KiB of float32 parameter values, excluding buffers/container metadata). Local recorded dual seed 42 CPU batch-one median inference was 1.48 ms over 20 repetitions. This is a development computer timing, not a Raspberry Pi benchmark. PairSE can cache all reference embeddings so reference signals need not be re-encoded on every update; latency and memory on the final device still require measurement.

VII. CONCLUSION

V2 delivers two compact population input modes and an explicit, reference-conditioned personal model. The six-person dual-input result demonstrates simultaneous improvement over personal mean and additive bias correction across seeds. Expanded evaluation confirms improvement over personal mean, while identifying bias correction as a strong DBP comparator. The final package preserves both findings and supplies the models needed to reproduce them. The next scientifically useful comparison is a larger personal development cohort with a newly reserved target cohort, followed by separately evaluated ND sensor/cuff transfer. The present results estimate scalar current BP from observed signals; they are neither future-ABP-waveform forecasts nor clinical-device validation.

REFERENCES

[1] W. Wang, P. Mohseni, K. L. Kilgore, and L. Najafizadeh, "PulseDB: A large, cleaned dataset based on MIMIC-III and VitalDB for benchmarking cuff-less blood pressure estimation methods," *Frontiers in Digital Health*, vol. 4, p. 1090854, 2023. [Online]. Available: https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2022.1090854/full

[2] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7132–7141. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2018/html/Hu_Squeeze-and-Excitation_Networks_CVPR_2018_paper.html

[3] O. Schlesinger, N. Vigderhouse, Y. Moshe, and D. Eytan, "Estimation and tracking of blood pressure using routinely acquired photoplethysmographic signals and deep neural networks," *Critical Care Explorations*, vol. 2, no. 4, p. e0095, 2020. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC7188414/

[4] J. Leitner, P.-H. Chiang, and S. Dey, "Personalized blood pressure estimation using photoplethysmography: A transfer learning approach," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 1, pp. 218–228, 2022. [Online]. Available: https://doi.org/10.1109/JBHI.2021.3085526

TABLE V

POPULATION POOLED METRICS, AVERAGED ACROSS THREE SEEDS. ALL PRESSURE PAIRS ARE SBP / DBP. THRESHOLDS ARE ABSOLUTE ERRORS IN MMHG; SD USES N-1. BOLD: BEST PER PRESSURE WITHIN EACH SPLIT.

Split / input	MAE	RMSE	Bias	SD	Within 5 (%)	Within 10 (%)	Within 15 (%)
validation / dual	12.51 / **8.26**	15.95 / **10.55**	**-1.19** / **-0.03**	15.90 / **10.54**	**26.13** / **37.91**	**48.54** / **67.08**	66.09 / **85.17**
validation / ppg	**12.50** / 8.41	**15.92** / 10.71	-1.43 / -0.72	**15.83** / 10.62	25.39 / 37.28	48.35 / 66.37	**66.64** / 84.30
test / dual	12.39 / 8.00	15.89 / 10.19	**-1.76** / **-0.89**	15.78 / 10.15	**26.03** / **38.83**	**49.22** / 68.91	67.60 / 86.68
test / ppg	**12.32** / **7.85**	**15.73** / **9.88**	-2.05 / -1.64	**15.57** / **9.67**	25.52 / 38.39	48.67 / **69.38**	**67.82** / **87.85**
aami / dual	**19.46** / **12.42**	**26.33** / **16.14**	**-13.22** / **-8.21**	**22.76** / 13.90	17.82 / **25.23**	**35.79** / **49.65**	**51.75** / **69.92**
aami / ppg	20.10 / 12.99	27.16 / 16.47	-13.95 / -9.11	23.30 / **13.69**	**19.22** / 22.87	35.59 / 45.80	50.35 / 67.17

Table notation: bold marks the best value per SBP/DBP in the stated comparison group: lower MAE, RMSE and SD, Bias closest to zero, and higher within-threshold percentages. Comparisons use unrounded values; numerical ties are all bold. Bold does not imply statistical significance or determine model selection.

TABLE VI

ORIGINAL SIX PEOPLE: MACRO METRICS, THREE-SEED AVERAGE. PAIRS ARE SBP / DBP; ERRORS IN MMHG. MACRO SD AVERAGES WITHIN-PERSON SAMPLE SD. BOLD: BEST PER PRESSURE ACROSS ALL SIX ROWS.

Split / input	MAE	RMSE	Bias	SD	Within 5 (%)	Within 10 (%)	Within 15 (%)
dual / Mean	9.47 / 4.22	12.19 / 5.46	**0.50** / -0.95	11.14 / 4.82	35.72 / 68.34	64.56 / 92.32	80.14 / 97.75
dual / Bias correction	9.16 / 4.40	11.23 / 5.27	0.60 / **-0.46**	9.35 / 4.25	33.26 / 62.42	63.41 / 93.89	81.90 / **99.53**
dual / PairSE	8.32 / **3.76**	10.27 / **4.69**	0.98 / -0.53	9.07 / **4.11**	36.65 / **71.51**	68.06 / **95.53**	86.12 / 99.28
ppg / Mean	9.47 / 4.22	12.19 / 5.46	**0.50** / -0.95	11.14 / 4.82	35.72 / 68.34	64.56 / 92.32	80.14 / 97.75
ppg / Bias correction	8.27 / 4.28	10.31 / 5.34	-1.49 / -1.83	**8.49** / 4.15	**39.01** / 63.42	**68.40** / 93.20	85.32 / 99.11
ppg / PairSE	**8.19** / 3.85	**10.17** / 4.83	-1.89 / -1.79	8.66 / 4.26	37.68 / 70.85	68.21 / 94.91	**86.24** / 99.11

TABLE VII

SUBSEQUENT SIXTEEN PEOPLE: MACRO METRICS, THREE-SEED AVERAGE. PAIRS ARE SBP / DBP; ERRORS IN MMHG. MACRO SD AVERAGES WITHIN-PERSON SAMPLE SD. BOLD: BEST PER PRESSURE ACROSS ALL SIX ROWS.

Split / input	MAE	RMSE	Bias	SD	Within 5 (%)	Within 10 (%)	Within 15 (%)
dual / Mean	17.41 / 9.71	19.56 / 10.87	3.23 / 3.02	11.18 / 5.73	15.06 / 23.73	28.68 / 57.18	42.71 / 81.01
dual / Bias correction	12.82 / 7.48	14.60 / 8.56	5.19 / 3.11	8.38 / 4.83	21.87 / 40.94	42.33 / 69.61	61.71 / 91.21
dual / PairSE	12.67 / 7.61	14.58 / 8.72	2.93 / 1.78	9.18 / 4.92	**27.74** / 40.67	**48.45** / 72.59	62.60 / 89.10
ppg / Mean	17.41 / 9.71	19.56 / 10.87	3.23 / 3.02	11.18 / 5.73	15.06 / 23.73	28.68 / 57.18	42.71 / 81.01
ppg / Bias correction	12.74 / **6.89**	14.33 / **7.95**	4.55 / 2.73	**8.07** / **4.57**	20.55 / **42.05**	40.49 / **76.80**	63.05 / **93.28**
ppg / PairSE	**12.47** / 7.51	**14.07** / 8.60	**2.63** / **1.61**	8.78 / 4.81	26.39 / 38.33	46.45 / 70.90	**63.59** / 88.89

TABLE VIII

EVERY PERSONAL SEED: PARTICIPANT-MACRO MAE (SBP / DBP), MMHG. THE REFERENCE-ONLY MEAN IS DETERMINISTIC, NOT THREE INDEPENDENT REPLICATIONS. BOLD: BEST METHOD PER PRESSURE WITHIN EACH ROW.

Cohort	Input	Seed	Personal mean	Bias correction	PairSE
Six	dual	42	9.47 / 4.22	9.00 / 4.35	**7.59** / **3.73**
Six	dual	43	9.47 / 4.22	9.91 / 4.66	**9.05** / **3.96**
Six	dual	44	9.47 / 4.22	8.58 / 4.20	**8.32** / **3.58**
Six	ppg	42	9.47 / 4.22	**7.85** / 4.37	8.07 / **3.95**
Six	ppg	43	9.47 / 4.22	**8.57** / 4.22	8.63 / **4.04**
Six	ppg	44	9.47 / 4.22	8.40 / 4.24	**7.88** / **3.55**
Sixteen	dual	42	17.41 / 9.71	**12.68** / **7.49**	12.78 / 7.73
Sixteen	dual	43	17.41 / 9.71	12.78 / 7.57	**12.62** / **7.50**
Sixteen	dual	44	17.41 / 9.71	13.00 / **7.36**	**12.62** / 7.60
Sixteen	ppg	42	17.41 / 9.71	12.68 / **6.85**	**12.40** / 7.29
Sixteen	ppg	43	17.41 / 9.71	12.91 / **6.94**	**12.42** / 7.68
Sixteen	ppg	44	17.41 / 9.71	12.62 / **6.87**	**12.58** / 7.57

TABLE IX

ORIGINAL SIX PEOPLE: POOLED METRICS, THREE-SEED AVERAGE. PAIRS ARE SBP / DBP; ERRORS IN MMHG. POOLED SD INCLUDES BETWEEN-PERSON OFFSETS. BOLD: BEST PER PRESSURE ACROSS ALL SIX ROWS.

Split / input	MAE	RMSE	Bias	SD	Within 5 (%)	Within 10 (%)	Within 15 (%)
dual / Mean	9.86 / 4.17	12.98 / 5.57	1.55 / -0.70	12.89 / 5.53	35.46 / 68.81	62.28 / 92.46	77.64 / 97.74
dual / Bias correction	9.61 / 4.57	12.11 / 5.66	2.20 / 0.20	11.88 / 5.61	32.15 / 60.35	61.21 / 93.11	79.67 / **99.39**
dual / PairSE	8.92 / **3.91**	11.52 / **5.13**	2.12 / **-0.11**	11.31 / 5.10	35.23 / 69.72	65.39 / **94.78**	83.23 / 99.13
ppg / Mean	9.86 / 4.17	12.98 / 5.57	1.55 / -0.70	12.89 / 5.53	35.46 / 68.81	62.28 / 92.46	77.64 / 97.74
ppg / Bias correction	8.51 / 4.17	10.99 / 5.39	**0.06** / -1.38	10.99 / 5.21	**38.10** / 64.92	**66.80** / 93.88	83.85 / 99.09
ppg / PairSE	**8.47** / 3.94	**10.80** / 5.16	-1.14 / -1.73	**10.73** / **4.86**	36.35 / **70.15**	66.69 / 94.49	**85.08** / 98.93

TABLE X

SUBSEQUENT SIXTEEN PEOPLE: POOLED METRICS, THREE-SEED AVERAGE. PAIRS ARE SBP / DBP; ERRORS IN MMHG. POOLED SD INCLUDES BETWEEN-PERSON OFFSETS. BOLD: BEST PER PRESSURE ACROSS ALL SIX ROWS.

Split / input	MAE	RMSE	Bias	SD	Within 5 (%)	Within 10 (%)	Within 15 (%)
dual / Mean	16.96 / 9.34	20.53 / 10.99	0.39 / 1.33	20.53 / 10.91	16.80 / 24.24	32.25 / 59.61	48.18 / 83.07
dual / Bias correction	**11.77** / 6.83	**14.68** / 8.79	3.00 / 1.64	**14.36** / 8.63	27.50 / 45.63	**48.69** / 75.11	**67.60** / 92.73
dual / PairSE	13.01 / 7.66	16.57 / 9.73	0.58 / 0.67	16.57 / 9.71	**28.04** / 41.79	47.62 / 70.39	61.95 / 87.94
ppg / Mean	16.96 / 9.34	20.53 / 10.99	0.39 / 1.33	20.53 / 10.91	16.80 / 24.24	32.25 / 59.61	48.18 / 83.07
ppg / Bias correction	12.03 / **6.58**	14.74 / **8.25**	2.72 / 1.92	14.49 / **8.03**	26.22 / **46.35**	47.40 / **76.83**	65.08 / **94.13**
ppg / PairSE	12.34 / 7.30	15.29 / 9.16	**-0.25** / **0.37**	15.28 / 9.15	26.25 / 42.51	47.55 / 71.73	65.05 / 89.68